TANVIRUL ISLAM

SOFTWARE ENGINEER

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Portfolio | LinkedIn | GitHub

SUMMARY

Backend engineer specialising in scalable, secure systems with **Java** and **Spring Boot**. Currently building a payment gateway (PSO) made of 20+ microservices, covering card, MFS and net-banking payments, card-data encryption, signed merchant webhooks and multi-channel notifications. Previously built a multi-tenant OTA platform serving three travel brands. Strong in RESTful API design, event-driven architecture, database optimisation and caching and search, and in turning business requirements into reliable, maintainable systems and resolving complex production issues.

TECHNICAL SKILLS

Languages: Java, SQL, JavaScript, TypeScript, C/C++
Backend: Spring Boot, Spring Data JPA, RESTful APIs, Microservices, Kafka, Batch Processing
Data & Search: MySQL, PostgreSQL, SQL Server, Redis (standalone/cluster), Meilisearch, Flyway, Liquibase, MinIO
Payments & Security: Card / MFS / net-banking integration, ECDH, HKDF, AES-GCM, HMAC signing, TOTP, key management, webhooks
Reporting: JasperReports
Tools: Git, GitLab CI, Postman, Playwright, TestNG, Log4j2, Grafana, Linux

PROFESSIONAL EXPERIENCE

PayNext | **Software Engineer** **Aug 2025 - Present**
paynext.com.bd - *Payment Systems Operator platform (current project)*

- Build and maintain a broad set of **Java 25 / Spring Boot 4 microservices** for a Payment Systems Operator (PSO) platform, covering checkout, processing, integration, card, notification, admin and merchant services.
- Developed the **Payment Checkout** and **Payment Processing** services that receive payment requests, create and validate orders, and drive the end-to-end payment lifecycle.
- Implemented the **Integration** service for banks and Mobile Financial Services: Visa, Mastercard and AMEX through acquirer banks, **bKash**, and **City Touch** net banking.
- Engineered **Card Services** with a per-session **ECDH** key. Card data is encrypted in the browser (HKDF + AES-GCM) and decrypted only inside the card vault, in line with end-to-end encryption policy.
- Centralised merchant API **key management (KMS)** and built **HMAC-signed merchant webhooks (IPN)** that support credential rotation, with one final, retry-safe notification per order.
- Restructured the **Notification Service** into email, SMS, push (**FCM**) and webhook channels, with category-based provider chains, ordered failover and **TOTP**-based OTP flows.
- Built versioned **TDR/MDR rate** management with effective windows, BIN-range handling and **multi-store** merchants across the admin and core services.
- Use **Kafka** for event-driven communication and **batch processing** for high-volume settlement and reconciliation, with **Redis** (standalone and cluster) for caching and **Flyway**-managed MySQL schemas.
- Contribute to the **Admin and Merchant** modules and an embeddable checkout SDK, supporting merchant setup, configuration and pre- and post-payment workflows.

TechnoNext Software Limited | **Software Engineer** **Nov 2023 - July 2025**
FirstTrip | Triplover | TakeOff Travels - *multi-tenant OTA platform*

- Developed scalable **Spring Boot** services for a **multi-tenant Online Travel Agency (OTA)** platform covering Hotel, Tour and Visa, serving FirstTrip, Triplover and TakeOff Travels from one codebase.
- Designed secure, high-performance **RESTful APIs** for booking flows, transaction processing and third-party supplier integrations, including the TBO hotel API.
- Implemented event-driven workflows with **Kafka** for payments, bookings and notifications.
- Managed the end-to-end payment lifecycle, including validation, processing and external gateway integration.
- Built and enhanced OTA modules such as search, booking management and pricing.
- Improved performance with **Redis** caching and optimised **PostgreSQL** queries on high-volume transactional data, and managed schema changes with **Liquibase**.
- Integrated **Meilisearch** to deliver fast, relevant search across travel data.
- Developed **tenant-based reporting** with **JasperReports**: branded invoices, refund invoices and business reports for stakeholders.
- Resolved critical production issues and contributed to scalable, fault-tolerant design across distributed services.

naztech Inc. | **Trainee Software Engineer** **Aug 2023 - Oct 2023**
Transaction Monitoring System / One Bank Limited

- Contributed to the design and optimisation of **RESTful APIs** for reliable service communication.
- Managed a **SQL Server** database, focusing on query optimisation and schema design.

EDUCATION

BSc. in Software Engineering **Jan 2019 - Mar 2023**
Daffodil International University *CGPA: 3.97/4.00*

Higher Secondary Certificate **2017**
Government Science College, Dhaka *GPA: 5.00/5.00*

Secondary School Certificate **2015**
Government Muslim High School, Chittagong *GPA: 5.00/5.00*

HONORS & AWARDS

- Dean's Award.
- PMI Kickoff Certification.

REFERENCES

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